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DEVICES AND METHODS FOR IMPROVED REMOVAL OF CORTICAL MATERIAL

Abstract

Devices and methods are disclosed for improved removal of cortical material. In certain embodiments, an ophthalmic surgical instrument includes a shaft having a distal end configured to be inserted into an eye of a patient toward a treatment area, and a projection located at the distal end of the shaft, the projection having a distal surface that extends at an angle to a longitudinal axis of the shaft, the distal surface configured to be positioned facing the treatment area during use of the ophthalmic surgical instrument. The projection is located on a tip or a sleeve of the ophthalmic surgical instrument.

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Background/Summary

INTRODUCTION

[0001] Cataract surgery involves removing a cataractous lens and replacing the lens with an artificial intraocular lens (IOL). The cataractous lens is typically removed by fragmenting the lens and aspirating the lens fragments out of the eye. The lens may be fragmented using, e.g., a phacoemulsification probe, a laser probe, or another suitable instrument. After the lens has been fragmented, a secondary irrigation/aspiration (I/A) handpiece and or a tip may be used to aspirate the fragments out of the eye, and to polish the lens capsule. The secondary I/A handpiece may also be used to irrigate fluid into the eye to maintain an intraocular pressure (IOP) and prevent collapse of the eye. A surgeon may therefore need to remove the phacoemulsification probe from the eye to switch fluidic lines between the phacoemulsification probe and the secondary I/A handpiece during the procedure to remove the fragmented lens and maintain stability of the eye's intraocular pressure (IOP) and anterior chamber (AC).

[0002] However, inserting and removing different ophthalmic surgical instruments throughout the course of a procedure can result in unwanted and unintentional trauma to ocular tissues, which can potentially lead to other ophthalmic complications. Further, having to switch between different ophthalmic surgical instruments throughout the procedure may increase the amount of time needed to complete the procedure, which further increases patient risk.

BRIEF SUMMARY

[0003] The present disclosure relates generally to ophthalmic surgical instruments for removal of cortical material.

[0004] In certain embodiments, an ophthalmic surgical instrument is provided. The ophthalmic surgical instrument includes a shaft having a distal end configured to be inserted into an eye of a patient toward a treatment area, and a projection located at the distal end of the shaft, the projection having a distal surface that extends at an angle to a longitudinal axis of the shaft, the distal surface configured to be positioned facing the treatment area during use of the ophthalmic surgical instrument.

[0005] In certain embodiments, a method of performing an ophthalmic procedure is provided. The method includes inserting a distal end of an ophthalmic surgical instrument into an eye of a patient toward a treatment area, the ophthalmic surgical instrument comprising a shaft and a projection, performing a phacoemulsification procedure with the ophthalmic surgical instrument in the eye, and without withdrawing the distal end of the ophthalmic surgical instrument from the eye, performing capsule polishing with the projection.

[0006] The following description and the related drawings set forth in detail certain illustrative features of one or more embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] The appended figures depict certain aspects of the one or more embodiments and are therefore not to be considered limiting of the scope of this disclosure.

[0008] FIG. 1A shows an example ophthalmic surgical system that may be used to perform ophthalmic procedures on an eye, according to certain embodiments.

[0009] FIG. 1B shows example components of a surgical console of the ophthalmic surgical system shown in FIG. 1A, according to certain embodiments.

[0010] FIG. 2 shows an example of an ophthalmic surgical instrument with a laser optical fiber, according to certain embodiments.

[0011] FIG. 3 shows an enlarged view of a distal end of the ophthalmic surgical instrument of FIG. 2, according to certain embodiments.

[0012] FIG. 4 shows a tip that can be used in conjunction with the ophthalmic surgical instrument of FIG. 2, according to certain embodiments.

[0013] FIG. 5 shows the distal end of an ophthalmic surgical instrument with the tip of FIG. 4, according to certain embodiments.

[0014] FIG. 6 shows another view of the distal end of the ophthalmic surgical instrument of FIG. 5, according to certain embodiments.

[0015] FIG. 7A shows an irrigation sleeve for the ophthalmic surgical instrument shown in FIG. 2, according to certain embodiments.

[0016] FIG. 7B shows the ophthalmic surgical instrument of FIG. 2 with the sleeve shown in FIG. 7A, according to certain embodiments.

[0017] FIG. 8A shows a distal end of an ophthalmic surgical instrument with an optical fiber disposed outside the shaft, according to certain embodiments.

[0018] FIG. 8B is a cross-sectional side view of the ophthalmic surgical instrument shown in FIG. 8A, according to certain embodiments.

[0019] FIGS. 9A-9B show the ophthalmic surgical instrument of FIGS. 8A-8B being inserted into an eye, according to certain embodiments.

[0020] FIG. 9C shows the ophthalmic surgical instrument of FIGS. 8A-8B inserted through a capsulotomy in the capsular bag of the eye shown in FIGS. 9A-9B, according to certain embodiments.

[0021] FIG. 10A shows the distal end of the ophthalmic surgical instrument shown in FIGS. 8A-8B with a tip having grooves, according to certain embodiments.

[0022] FIG. 10B shows the distal end of the ophthalmic surgical instrument shown in FIGS. 8A-8B with a tip having notches, according to certain embodiments.

[0023] FIG. 10C shows the distal end of the ophthalmic surgical instrument shown in FIGS. 8A-8B with a tip having wings, according to certain embodiments.

[0024] FIG. 10D shows the distal end of the ophthalmic surgical instrument shown in FIGS. 8A-8B with a tip having a textured surface, according to certain embodiments.

[0025] FIG. 11A shows a distal end of an ophthalmic surgical instrument with a tip, according to certain embodiments.

[0026] FIG. 11B shows the distal end of the ophthalmic surgical instrument with another tip, according to certain embodiments.

[0027] FIG. 12 illustrates a flowchart of a method for performing an ophthalmic surgical procedure, according to certain embodiments.

[0028] To facilitate understanding, identical reference numerals have been used, where possible, to designate identical elements that are common to the drawings. It is contemplated that elements and features of one embodiment may be beneficially incorporated in other embodiments without further recitation.

DETAILED DESCRIPTION

[0029] It will be readily understood that the components of the embodiments as generally described herein and illustrated in the appended Figures can be arranged and designed in a wide variety of different configurations. Thus, the following more detailed description of various embodiments, as represented in the Figures, is not intended to limit the scope of the present disclosure but is merely representative of various embodiments. While the various aspects of the embodiments are presented in the Figures, the Figures are not necessarily drawn to scale unless specifically indicated.

[0030] Reference throughout this specification to the term “distal” refers to a system, device, component, end, portion, or segment that is disposed closer to a patient and/or further from a console during an ophthalmic procedure; and the term “proximal” refers to the system, device,

component, end, portion, or segment that is disposed further from the patient and/or closer to the console during the ophthalmic procedure.

[0031] Current cataract surgeries often employ an ophthalmic surgical instrument (e.g., a phacoemulsification probe, a laser probe, or another suitable instrument) that is used to fragment a cataractous lens along with another ophthalmic surgical instrument (e.g., a secondary irrigation/aspiration (I/A) handpiece) for aspirating the fragmented lens and irrigating fluid (e.g., a balanced salt solution (BSS) or other irrigation solution) into the eye. As such, a surgeon may need to switch between the different ophthalmic surgical instruments one or more times throughout the surgery, as the surgeon can typically only hold two instruments. However, exchanging different instruments into and out of the eye throughout the surgery increases the risk of infection and the likelihood of unwanted and unintentional trauma to ocular tissues, and may also increase a duration of the surgery, which further increases risk to the patient.

[0032] Further, after the cataractous lens has been fragmented, a capsular wall of the lens may need to be polished to ensure sufficient removal of residual lens epithelial cells. However, due to a small surface area and rounded distal end, current ophthalmic surgical instruments struggle to remove the residual lens epithelial cells.

[0033] Accordingly, the embodiments described herein provide multifunctional instruments and a variety of tips that can be used with different ophthalmic surgical instruments, and which include features for the removal of residual lens epithelial cells. For example, when used with an ophthalmic surgical instrument, each of the tips herein enables the ophthalmic surgical instrument to be multifunctional, such that the ophthalmic surgical instrument can perform a certain surgical functionality (e.g., fragmentation, aspiration, and/or irrigation) while also being able to remove lens epithelial cells during capsule polishing. The multifunctional instruments described herein, therefore, reduce the need to switch between different instruments throughout the surgery, the likelihood of unwanted and unintentional trauma to ocular tissues, and the duration of the surgery.

[0034] FIG. 1A shows an example ophthalmic surgical system **100** that may be used to perform ophthalmic procedures on an eye, according to certain embodiments. The ophthalmic surgical system **100** includes a console **102** (also referred to as a “surgical console”), which includes a display **104**, an input device **106** (e.g., a foot pedal), and an ophthalmic surgical instrument **101** (also referred to as an “instrument” or a “handpiece”). The components of the ophthalmic surgical system **100** and the surgical console **102** are mechanically and/or electrically coupled as shown and described in more detail with reference to FIG. 1B.

[0035] The surgical console **102** may be similar to surgical consoles as shown and described in U.S. Pat. No. 9,931,447, the entire disclosure of which is hereby expressly incorporated herein by reference. The surgical console **102** may be similar to surgical consoles that have been known and used, such as the CENTURION® Vision System available from Alcon Laboratories, Inc. (Fort Worth, Texas) or the CONSTELLATION® Vision System available from Alcon Laboratories, Inc. (Fort Worth, Texas), or any other ophthalmic surgical console suitable for use with the principles described herein.

[0036] FIG. 1B shows example components of the surgical console **102** of the ophthalmic surgical system **100** shown in FIG. 1A, according to certain embodiments. As shown, the surgical console **102** includes a controller **112**, an input subsystem **114**, a handpiece subsystem **116**, an aspiration subsystem **118**, an irrigation subsystem **130**, and a display **104**. The controller **112** controls the operation of the surgical console **102** and is illustrated as being operationally coupled by a wired or wireless connection to the input device **106** via input subsystem **114**, and to the ophthalmic surgical instrument **101** via handpiece subsystem **116**, aspiration subsystem **118**, irrigation subsystem **130**, and a laser subsystem **132**. The controller **112** includes a processor **120**, a memory **122**, and controller circuitry **124**.

[0037] Processor **120** may be any type of general purpose processor or could be a processor specifically designed for driving the subsystems illustrated in FIG. 1B, such as an application-

specific integrated circuit (“ASIC”). The processor **120** may be, or include, a microprocessor, a microcontroller, an embedded microcontroller, a programmable digital signal processor, or any other programmable device operable to execute instructions stored in the memory **122** for operating surgical console **102**. For example, the processor **120** may execute instructions in the memory **122** to receive inputs provided by input device **106** through input subsystem **114** and, in response, send instructions to the aspiration subsystem **118** and/or the handpiece subsystem **116** for operating the ophthalmic surgical instrument **101**. Further, the processor **120** may execute instructions to generate a user interface view for display by display **104**. In some instances, the processor **120** may also be or include a programmable gate array, programmable array logic, or any other device of combinations of devices operable to process electric signals.

[0038] Memory **122** can be any type of storage device or non-transitory computer-readable medium, such as random-access memory (“RAM”) or read-only memory (“ROM”), which is operable to receive, store, or recall data, including, but not limited to, electronic, magnetic, or optical memory, whether volatile or non-volatile. The memory **122** stores instructions executed by the processor **120**. In example embodiments, functionality disclosed herein can be provided by the processor **120** and the memory **122** (i.e., software based), by the controller circuitry **124** (i.e., hardware based), or by a combination thereof. The memory **122** may include code stored thereon. The code may include instructions that may be executable by the processor **120**. The code may be created, for example, using any programming language, including but not limited to, C, C++, Java, Python, Rust, or any other programming language (including assembly languages, hardware description languages, and database programming languages). In some instances, the code may be a program that, when executed by the processor **120**, causes the surgical console **102** to operate subsystems **114**, **116**, **118**, **130** and/or **132** for, e.g., driving the ophthalmic surgical instrument **101** or other devices in communication with the surgical console **102**.

[0039] Ophthalmic surgical instrument **101** may be any suitable ophthalmic surgical instrument that can be operated on the basis of the embodiments described herein. For example ophthalmic surgical instrument **101** may be a phacoemulsification (phaco) handpiece (also referred to as a “phaco probe”). Other examples of the ophthalmic surgical instrument **101** may include a laser probe instrument, an irrigation instrument, an aspiration instrument, or other similar probes.

[0040] Handpiece subsystem **116** is configured to facilitate the operation of the ophthalmic surgical instrument **101**. For example, handpiece subsystem **116** may control the operations (e.g., activation and deactivation) of a component of the ophthalmic surgical instrument **101**.

[0041] Aspiration subsystem **118** provides aspiration control for ophthalmic surgical instrument **101**. In some embodiments, aspiration subsystem **118** may be operatively coupled to a surgical cassette during a surgical procedure. For example, the surgical cassette may be inserted into, attached to, and/or integrated with aspiration subsystem **118** via a coupling mechanism. When aspiration subsystem **118** is operatively coupled to a surgical cassette, aspiration subsystem **118** may control aspiration through the surgical cassette, which may in turn be coupled to the ophthalmic surgical instrument **101**, for example, via a cable or other tether. In certain embodiments, the aspiration subsystem **118** includes one or more mechanical pumps having roller pump heads configured to engage with one or more corresponding pump assemblies on the surgical cassette. The engagement of the roller pump heads and pump assemblies generates a source of pressure and/or vacuum utilized during an ophthalmic surgical procedure.

[0042] Irrigation subsystem **130** provides irrigation control for ophthalmic surgical instrument **101**. In some embodiments, irrigation subsystem **130** may be operatively coupled to a fluid bag during a surgical procedure. For example, the fluid bag may be inserted into, attached to, and/or integrated with irrigation subsystem **130** via a coupling mechanism. When irrigation subsystem **130** is operatively coupled to a fluid bag, irrigation subsystem **130** may control irrigation through the fluid bag, which may in turn be coupled to the ophthalmic surgical instrument **101**, for example, via a cable or other tether. In certain embodiments, the irrigation subsystem **130** includes one or more

mechanical plates configured to engage with the fluid bag. The engagement of the fluid bag generates a source of pressure utilized to irrigate fluid during an ophthalmic surgical procedure. [0043] Laser subsystem **132** may comprise a laser that may be housed within the surgical console **102** or elsewhere, such as in a separate console that communicates with the surgical console **102**. The laser subsystem **132** may include components for operating the laser, such as a power supply, controller, laser pumps, laser energy control, and/or monitor. In addition, the laser subsystem **132** may include components in the optical path of the laser output, such as one or more lenses, mirrors, and/or optical fibers.

[0044] In some embodiments, the laser subsystem **132** may be suitable for cataract surgery. In some embodiments, the output energy of the laser subsystem **132** is suitable for fragmentation and/or emulsification of a cataractous lens. In some examples, the laser output is used for fragmentation and/or emulsification of the lens to a sufficient degree for removal of the lens.

[0045] The laser may be any type of laser suitable for the desired application. The laser may output suitable electromagnetic radiation at any suitable wavelength. For example, the laser may emit electromagnetic radiation in one or more wavelengths in the visible, infrared, and/or ultraviolet wavelengths. The laser may operate or be operated to emit a continuous beam of electromagnetic radiation. Alternatively, the laser may operate or be operated to emit a pulsed beam.

[0046] In one example, the laser operates in the infrared range. For example, the laser may output electromagnetic radiation in the mid-infrared range, for example in a wavelength range of about 2.0 microns to about 4.0 microns. Some example wavelengths include about 2.5 microns to about 3.5 microns, such as about 2.775 microns, about 2.8 microns, or about 3.0 microns. Such a laser may be suitable, for example, for lens fragmentation or emulsification in cataract surgery, or for other procedures.

[0047] The laser subsystem **132** is designed to direct the laser electromagnetic radiation from the laser to an output port. The laser subsystem **132** may direct the laser electromagnetic radiation from the laser to the output port through one or more optical components, such as lenses and mirrors.

[0048] The ophthalmic surgical instrument **101** may be optically connected to the laser subsystem **132** to receive the laser electromagnetic radiation from the output port. The ophthalmic surgical instrument **101** may be connected to the laser subsystem **132**, e.g., by a cable with an optical fiber. The connection (e.g., cable) may be flexible and relatively long to give the operator flexibility in maneuvering the ophthalmic surgical instrument **101** at some distance away from the laser subsystem **132**. The laser electromagnetic radiation may be transmitted to the ophthalmic surgical instrument **101** from the laser subsystem **132** through the optical fiber of the connecting cable. The optical fiber of the connecting cable may continue through the ophthalmic surgical instrument **101** or may be optically connected to an optical fiber in the ophthalmic surgical instrument **101**. In either case, the ophthalmic surgical instrument **101** may include an optical fiber that terminates at the distal end of the ophthalmic surgical instrument **101**. The optical fiber may carry the laser electromagnetic energy from the laser and emit it from the distal end of the optical fiber at the distal end of the ophthalmic surgical instrument **101** to the desired target, such as a lens or lens fragment in the eye of a patient.

[0049] Input device **106** may be any device that is capable of receiving commands from the user of the surgical console **102** in order to operate the ophthalmic surgical instrument **101** and/or other components of the surgical console **102**. In FIG. 1A, input device **106** is illustrated as a foot pedal, however, other types of input devices are also within the scope of the disclosure. In one example, the user provides a command to the input device **106**, which is received and relayed to the controller **112** by the input subsystem **114**. In response, the controller **112** sends instructions to the handpiece subsystem **116**, aspiration subsystem **118**, and/or irrigation subsystem **130** to control the operations of the ophthalmic surgical instrument **101** based on the user command.

[0050] FIG. 2 shows an example of the ophthalmic surgical instrument **101** with a laser optical fiber **240** (e.g., a sapphire optical fiber), according to certain embodiments. FIG. 3 shows an

enlarged view of the distal end **202** of the ophthalmic surgical instrument **101** of FIG. 2, according to certain embodiments. Accordingly, FIGS. 2 and 3 are described together herein for clarity purposes.

[0051] The ophthalmic surgical instrument **101** comprises a housing **204** and a shaft **210**, e.g., a cannula, that extends from the distal end **212** of the housing **204**. The housing **204** may be hollow with an internal chamber in which internal components of the instrument **101** may be housed and protected. The housing **204** may have an external surface that can be grasped by an operator of the instrument **101**, such as a surgeon.

[0052] The shaft **210** may be tubular, e.g., a hollow tube such as a cannula, that has an aspiration port **214** at its distal end **212** which may be used for aspiration. For example, by applying suction through an aspiration channel from the proximal end of an aspiration luer at the proximal end of the housing **204**, fluid and/or tissue, such as lens or other tissue fragments, may be aspirated through the aspiration port **214** in the distal end **212** of the shaft **210**.

[0053] The optical fiber **240** may be located inside of the shaft **210**, as shown. The optical fiber tip **242** may be flush (in the same plane as) the end face **216** of the shaft **210**. In other embodiments, the optical fiber tip **242** may extend beyond the end face **216** of the shaft **210**, or the end face **216** of the shaft **210** may extend beyond the optical fiber tip **242**. The optical fiber tip **242** does not need to be inside of the aspiration port **214** but only in close proximity so that the material affected by the laser action can be readily aspirated. Examples of the optical fiber **240** located outside of the shaft **210** are shown in FIGS. 8A-8B, 9A-9B, and 10A-10D.

[0054] The ophthalmic surgical instrument **101** may also include an irrigation sleeve **220**. The irrigation sleeve **220** may serve to direct an irrigation fluid (e.g., saline) to the distal end **202** of the instrument **101**. The housing **204** may have an irrigation supply line through which an irrigation fluid may be introduced. The irrigation sleeve **220** may be coupled directly to the housing **204** or coupled to the housing **204** through another part of the ophthalmic surgical instrument **101**. In one example, the housing **204** may have external threads at its distal end, and the irrigation sleeve **220** may have internal threads at its proximal end, by which the irrigation sleeve **220** may be attached to the housing **204**. As can be seen in FIG. 2, the irrigation sleeve **220** generally has a proximal hub **226** having a relatively larger diameter for coupling to the housing **204** and a distal tube **228** in the shape of a narrow tube having a relatively smaller diameter for fitting around the shaft **210** at a small enough dimension for inserting through an incision in an eye.

[0055] In the illustrated example, the irrigation sleeve **220** is positioned around the shaft **210** to provide a fluid passageway or channel from the position of its attachment to the housing **204** through the space between the irrigation sleeve **220** and the shaft **210**. The irrigation sleeve **220** has an end opening **223** at its distal end **222** through which the distal end **212** of the shaft **210** extends and one or more side openings **224** at its distal end **222** adjacent to the end opening **223**. At the end opening **223**, the irrigation sleeve **220** has a snug fit around the shaft **210**. When an irrigation fluid is introduced through the supply line in the housing **204**, it passes through the channel of the irrigation sleeve **220** and out of the side opening(s) **224** at the distal end **222** of the irrigation sleeve **220**.

[0056] As an example, the distal tube **228** of the irrigation sleeve **220** may have an outer diameter in a range of 0.030 inches to 0.080 inches to fit through an incision in an eye and an inner diameter in a range of 0.020 inches to 0.070 inches to accommodate the shaft **210**. The end opening **223** at the distal end **222** of the irrigation sleeve **220** through which the distal end **212** of the shaft **210** projects may be a circular opening having a diameter in a range of 0.010 inches to 0.065 inches, which may be equal to or slightly smaller than the outer diameter of the distal end **212** of the shaft **210** to form a snug fit around the distal end **212** of the shaft **210**. The examples of dimensions and ranges of dimensions represent possible embodiments; other embodiments with different dimensions may be used.

[0057] The irrigation sleeve **220** may be made of an elastomeric material such as a compliant

silicone rubber. The irrigation sleeve **220** alternatively may be made of other materials, such as polyurethane, ethylene propylene, neoprene, or other suitable materials. An elastomeric material for the irrigation sleeve **220** allows some compliance and facilitates the snug fit between irrigation sleeve **220** and the shaft **210** at the area of the end opening **223**. An elastomeric material for the irrigation sleeve **220** also facilitates a seal or snug fit between the irrigation sleeve **220** and the adjacent eye tissue at the incision site, such as the cornea or sclera, which can help minimize leakage from the eye between the eye tissue and the irrigation sleeve **220**.

[0058] In the operation of the ophthalmic surgical instrument **101**, the operator inserts the distal end **202** of the ophthalmic surgical instrument **101** through a suitable incision in a patient's eye (e.g., as shown in FIGS. **9A-9B**) and positions the distal end **202** of the ophthalmic surgical instrument **101** at a desired location, such as adjacent to a cataractous lens of a patient. The distal end **212** of the shaft **210** is directed toward a treatment area, with the tip **242** at the distal end of the optical fiber **240** positioned facing the treatment area. Irrigation fluid such as saline may be supplied from the surgical console through the irrigation supply line and irrigation sleeve **220**, such that it flows out of the side opening(s) **224** to the target area. The laser subsystem may be activated to emit laser energy from the tip **242** of the optical fiber **240** (i.e., the distal end of the optical fiber **240**), which can break up and emulsify the desired tissue, such as the cataractous lens. At the same time, a pumping module may be used to apply suction through the shaft **210**, thereby suctioning through the aspiration port **214** fluid and tissue and/or lens fragments that have been separated by the action of the laser.

[0059] FIG. **4** shows a tip **450** for the ophthalmic surgical instrument **101**, according to certain embodiments. FIG. **5** shows the distal end **202** of the ophthalmic surgical instrument **101** with the tip **450** of FIG. **4**, according to certain embodiments. FIG. **6** shows another view of the distal end **202** of the ophthalmic surgical instrument **101** of FIG. **5**, according to certain embodiments. Accordingly, FIGS. **4**, **5**, and **6** are described together herein for clarity purposes.

[0060] The tip **450** has a cylindrical hub **452** and a projection **454**. In some embodiments, the projection may be one of the projections described in U.S. patent application Ser. No. 18/451,211 titled “Devices and Methods for Improved Followability in Laser-Based Ocular Procedures”, filed on Aug. 17, 2023, whose inventors are Francisco Javier Ochoa and John Morgan Bourne, which is hereby incorporated by reference in its entirety as though fully and completely set forth herein.

[0061] The cylindrical hub **452** is configured to fit over and on the distal end **212** of the shaft **210**, as can be seen in FIGS. **5** and **6**. The tip **450** has a center opening **458** that provides an entranceway to the aspiration port **214** (e.g., see FIG. **3**) of the shaft **210** for aspiration. The optical fiber **240** is located inside of the shaft **210** and inside of the center opening **458** of the tip **450**. In an alternative example, the tip **450** has a center opening **458** that provides an entranceway to the aspiration port **214** of the shaft **210** for aspiration, but the optical fiber **240** is located outside of the shaft **210**. For example, in such an embodiment, the optical fiber **240** may be located either inside or outside of the center opening **458** of the tip **450**. Locating the optical fiber **240** outside of the shaft **210** can help reduce clogging of the aspiration channel, by having the laser action offset from the aspiration channel.

[0062] When the tip **450** is coupled to the instrument **101** as shown in FIGS. **4**, **5**, and **6**, the projection **454** is located at the distal end **212** of the shaft **210** and at the distal end **202** of the instrument **101**. The projection **454** has a distal surface **456** in proximity to the distal tip **242** of the optical fiber **240**. Thus, the distal surface **456** of the projection **454** faces the treatment area when the optical fiber tip **242** faces the treatment area.

[0063] When the tip **450** is in use, the distal surface **456** provides instrument **101** a surface area facing the treatment area that is significantly larger than when the tip is not in use. For example, as shown in FIG. **3**, in the illustrated instrument **101** without the projection **454**, the surface area of instrument **101** facing the treatment area would be limited to the end face **216** of the shaft **210**, aside from the tip **242** of the optical fiber **240** itself. Therefore, the projection **454** provides a

significantly larger surface area facing the treatment area as compared to the end face **216** of the shaft **210**.

[0064] In one example, the end face **216** has an outer diameter of 0.0243 inches and an inner diameter of 0.0187 inches, with a surface area of 0.000198 square inches. In one example of the projection **454**, the surface area of the distal surface **456** facing the treatment area is 0.00164 square inches. Thus, in this example, the surface area of the distal surface **456** of the projection **454** is more than eight times greater than the surface area of the end face **216** of the shaft **210**.

[0065] In other examples, a surface area of the distal surface **456** may be at least two times greater than a surface area of the end face **216** of the shaft **210**, or at least four times greater than a surface area of the end face **216** of the shaft **210**, or at least six times greater than a surface area of the end face **216** of the shaft **210**. The distal surface **456** of the projection **454** may have a length (L1) of 0.020 inches to 0.090 inches and a width (W1) of 0.020 inches to 0.060 inches. Smaller or larger lengths and/or widths may be used. In one example, the distal surface **456** of the projection **454** has a length of about 0.050 inches and a width of about 0.020 inches. The distal surface **456** of the projection **454** may be dished or curved in whole or in part, thereby presenting a concave surface facing the treatment area, in better conformity to the shape of the bubbles induced by the laser action.

[0066] The projection **454**, with its distal surface **456** presenting a surface area that faces the area treated by the laser, helps inhibit the bubbles formed by the laser from moving away from the optical fiber tip **242**. Without the surface area of the projection **454**, the action of the laser and the bubble dynamics can, in certain situations, cause the bubbles formed by the laser to move away from the distal end of the instrument. This can reduce the efficiency of the laser action or otherwise detrimentally affect the performance of the instrument **101**.

[0067] Without a projection and associated surface area facing the treatment area, a system may have good holding power for high vacuum levels or low power settings, but at higher laser power settings the laser action and bubble dynamics may cause repulsion of the resultant bubbles. Accordingly, the cutting power or emulsification efficiency decreases significantly, since the target tissue or material is being moved away from the distal end of the instrument tip as opposed to broken up and aspirated into the cannula.

[0068] The use of a projection with a surface area facing the treatment area (e.g., surface area of distal surface **456**) helps inhibit the distal movement of the bubble action from the optical fiber tip **242**. At a minimum, it reduces the repulsive effects of the laser-induced bubble dynamics. In some embodiments, it can reverse the effect such that repulsion becomes attraction. In cataract surgical terms, the use of a projection with its surface area facing the treatment area improves followability.

[0069] In some embodiments, the tip **450** with the projection **454** is a plastic part that is added to the shaft **210** (e.g., a metal cannula). The projection **454** can also be shaped out of the shaft or cannula itself or integrated as part of the irrigation sleeve **220** (e.g., as shown in FIGS. 7A-7B). The shape of the projection **454** can have different configurations depending on the performance requirements.

[0070] In some embodiments, the tip **450**, including the hub **452** and the projection **454**, may be transparent, in order to facilitate visualization by the operator through the material. The tip **450** may be comprised of a polymeric material. As an example, the tip **450** may be comprised of polycarbonate.

[0071] In some embodiments, the projection **454** and the distal surface **456** may be asymmetrical about a plane along the longitudinal axis **218** of the shaft **210**. This asymmetry can allow a cutting or shaving action on one side of the instrument **101**.

[0072] In some embodiments, the projection **454** may be angled with respect to the longitudinal axis **218** of the shaft **210** (e.g., at an angle from 0° (degrees) to 90° (perpendicular) relative to the longitudinal axis **218**). This can help facilitate insertion of the distal end **202** of the instrument **101** into the eye (e.g., as shown in FIGS. 9A-9B). That is, the operator can tilt the instrument **101** at a

first angle to maneuver the projection **454** into the incision in the eye (e.g., as shown in FIG. **9A**), and then tilt the instrument **101** at a second angle to maneuver the shaft **210** through the incision in the eye (e.g., as shown in FIG. **9B**).

[0073] FIG. **7A** shows an irrigation sleeve **720** for the ophthalmic surgical instrument **101** shown in FIG. **2**, according to certain embodiments. FIG. **7B** shows the ophthalmic surgical instrument **101** and the sleeve **720** shown in FIG. **7A** coupled together, according to certain embodiments.

Accordingly, FIGS. **7A-7B** are described together herein for clarity purposes.

[0074] The irrigation sleeve **720** may be the same as the irrigation sleeve **220** described with reference to FIG. **2**, except that the sleeve **720** includes a tip **750** that incorporates a projection **754** with a distal surface **756** (instead of the tip **450** with the projection **454** and the distal surface **456**). In such embodiments, the projection **754** is part of the irrigation sleeve **720** and may be molded as part of its distal end **702**. The distal surface **756** functions similarly to the distal surface **456** of the tip **450**.

[0075] In this illustrated example, the irrigation sleeve **720** has a center opening **758** that provides an entranceway to the channel of the shaft **210** for aspiration and side openings **724** for fluid irrigation. Additionally, the optical fiber **240** is located inside of the shaft **210** and the center opening **758**. In an alternative example, the irrigation sleeve **720** again has a center opening that provides an entranceway to the channel of the shaft **210** for aspiration, but the optical fiber **240** is located outside of the shaft **210**, and the optical fiber **240** may be located either inside or outside of the center opening of the irrigation sleeve **720**.

[0076] The irrigation sleeve **720**, including the projection **754**, may be transparent, in order to facilitate visualization by the operator through the material. The projection **754** may be formed of the same material or a different material as the irrigation sleeve **220**, e.g., a polymeric material such as polycarbonate or an elastomeric material such as silicone rubber.

[0077] The projection **754** and the distal surface **756** may be asymmetrical about a plane along the longitudinal axis **218** of the shaft **210**. This can allow a cutting or shaving action on one side of the instrument **101**. Also, the projection **754** may be angled with respect to the longitudinal axis **218** of the shaft **210** (e.g., at an angle from 0° to 90° (perpendicular) relative to the longitudinal axis **218**), to facilitate insertion of the distal end **702** of the instrument **101** into the eye, as described above.

[0078] The irrigation sleeve **720** may be coupled directly to the housing **204** or coupled to the housing **204** through another part of the ophthalmic surgical instrument **101**. In one example, the housing **204** may have external threads at its distal end, and the irrigation sleeve **720** may have internal threads at its proximal end, by which the irrigation sleeve **720** may be attached to the housing **204**. In some examples, the irrigation sleeve **720** may be slideable relative to the housing **204**, such that the optical fiber **240** may be exposed and/or covered by moving the irrigation sleeve **720** longitudinally (e.g., along the longitudinal axis **218**) relative to the shaft **210**.

[0079] As can be seen in FIG. **7A**, the irrigation sleeve **720** has a proximal hub **726** configured to be coupled to the housing **204** and a distal tube **728** having a smaller diameter (relative to the diameter of the proximal hub **726**) for fitting around the shaft **210**. The sleeve **720** with projection **754** may be coupled to a commercially available phacoemulsification tip (e.g., a tip without a projection such as a balanced phacoemulsification tip or a hybrid phacoemulsification tip) to safely reach residual cortex under a capsulotomy with the projection **754**.

[0080] FIG. **8A** shows a distal end of an ophthalmic surgical instrument **801** with the optical fiber **240** disposed outside the shaft **210**, according to certain embodiments. FIG. **8B** is a cross-sectional side view of the ophthalmic surgical instrument **801** shown in FIG. **8A**, according to certain embodiments. Accordingly, FIGS. **8A-8B** are described together herein for clarity purposes.

[0081] In FIGS. **8A-8B**, the optical fiber **240** extends distally from a fiber tube **864** disposed adjacent to the shaft **210**, and through a tip **850** coupled to the shaft **210**. In particular, the optical fiber **240** is located outside of the shaft **210** and inside of a center opening **858** of the tip **850**. The fiber tube **864** may be laser welded to the shaft **210** and is configured to help prevent the optical

fiber **240** from breaking. In the illustrated example, an irrigation sleeve **820**, which is similar to the irrigation sleeve **220**, is positioned around the shaft **210** and the fiber tube **864**. The sleeve **820** includes two side openings **824** for irrigating fluid (e.g., BSS).

[0082] FIGS. **8A-8B** illustrate a tip **850** including a substantially flat distal surface **856**, a body **860** between cylindrical hub **852** and projection **854**, and a slot **862** disposed through the projection **854**. The tip **850** also includes a plurality of inner protrusions or projections **866** that are circumferentially arranged on the inner surface of the cylindrical hub **852** and extend inwards from the cylindrical hub **852** toward the longitudinal axis thereof. The inner projections **866** are configured to interact with or be positioned into corresponding apertures **868** similarly arranged on the outer surface of the distal end of the shaft **210**. By fitting into the corresponding apertures **868**, the inner projections **866** are configured to removably couple the tip **850** to the shaft **210**.

[0083] The projection **854** of the tip **850** has a thin cross-sectional profile (D) at its edge **855**. The thin cross-sectional profile (D) of the projection **854** may have a depth of 0.005 inches to 0.10 inches. The thin cross-sectional profile (D) of the projection **854** helps facilitate easier insertion of the ophthalmic surgical instrument **801** into the eye. Additionally, the projection **854** can be used as a hook to gently pull on a capsulotomy to further extend the reach of the ophthalmic surgical instrument **801**. For example, the projection **854** may be used to reach under the capsulotomy to facilitate removal of cortical material, including sub-incisional cortex, which may be particularly difficult to reach. Such uses of the ophthalmic surgical instrument **801** with the tip **850** are described in further detail with reference to FIGS. **9A-9C**.

[0084] The flat distal surface **856** of the tip **850** helps provide a lens capsule-friendly surface, which can be used to operate on and around the capsule membrane. In one example of the projection **854**, the surface area of the distal surface **856** facing the treatment area is 0.00164 square inches, which is more than eight times greater than the surface area of the end face **216** of the shaft **210**. In other examples, a surface area of the distal surface **856** of the projection **854** may be at least two times greater than a surface area of the end face **216** of the shaft **210**, or at least four times greater than a surface area of the end face **216** of the shaft **210**, or at least six times greater than a surface area of the end face **216** of the shaft **210**. The distal surface **856** of the projection **854** may have a length (L2) of 0.020 inches to 0.090 inches and a width (W2) of 0.020 inches to 0.060 inches. Smaller or larger lengths and/or widths may be used. In one example, the distal surface **856** of the projection **854** has a length of about 0.050 inches and a width of about 0.020 inches.

[0085] The optical fiber **240** is disposed through the slot **862** and is substantially flush with the distal surface **856**. In the illustrated example, the optical fiber **240** is positioned along an edge of the center opening **858**. By positioning the optical fiber **240** near an outer portion of the center opening **858**, flow of tissue and/or other material aspirated by the ophthalmic surgical instrument **801** through the center opening **858** is improved in comparison to the flow of materials through the center opening **458**. Accordingly, the positioning of the optical fiber **240** as shown in FIGS. **8A-8B** helps reduce clogging of the aspiration channel. Further, when the optical fiber tip **242** is flush with the distal surface **856**, the optical fiber **240** is less likely to break during use of the ophthalmic surgical instrument **801**.

[0086] The body **860** of the tip **850** defines an aspiration channel **870** therein which fluidly connects the center opening **858** to channel **215** of the shaft **210**. In the illustrated example, a longitudinal axis **819** of the body **860** defines an angle **821** relative to the longitudinal axis **218** of the shaft **210**. The angle **821** defined by the longitudinal axes **218**, **819** is between, for example, 95° and 175° (e.g., between 100° and 170°, 105° and 165°, 110° and 160°, or 115° and 155°). By angling the channel **870** of the tip **850** relative to the channel **215** of the shaft **210**, tissue and/or other materials aspirated by the ophthalmic surgical instrument **801** are guided away from the optical fiber **240** and are therefore less likely to get caught on and/or break the optical fiber **240**.

[0087] In some embodiments, the tip **850**, including the hub **852** and the projection **854**, may be transparent, in order to facilitate visualization by the operator through the material. The tip **850**

may be comprised of a polymeric material. As an example, the tip **850** may be comprised of polycarbonate. In some embodiments, the projection **854** can also be shaped out of the shaft or cannula itself, such that the projection **854** may be formed of the same material as the shaft or cannula. The shape of the projection **854** can have different configurations depending on the performance requirements.

[0088] Although the ophthalmic surgical instrument **801** is shown as including the optical fiber **240** with the optical fiber tip **242** extending to the distal surface **856** of the projection **854**, in some embodiments, the optical fiber tip **242** may not extend to the distal surface **856**, and in some other embodiments, the ophthalmic surgical instrument **801** may not include the optical fiber **240** (e.g., in an ultrasonic phacoemulsification tip).

[0089] FIGS. **9A-9B** show the ophthalmic surgical instrument **801** of FIGS. **8A-8B** being inserted into an eye **900**, according to certain embodiments. For brevity, reference numbers for the eye **900** are only shown for cornea **902**, outer surface **908** of the cornea **902**, an anterior segment **906** (e.g., an anterior chamber), an iris **912**, a lens **914**, and a capsular bag **916**.

[0090] In FIG. **9A**, the ophthalmic surgical instrument **801** is introduced into the eye **900** via a clear corneal incision, such as incision **904**. The incision **904** is configured to provide the ophthalmic surgical instrument **801** access to the anterior segment **906** of the eye **900**. To help with the insertion of the ophthalmic surgical instrument **801**, the ophthalmic instrument **801** is angled such that the projection **854** is first inserted into the incision **904**. The projection **854** facilitates easier insertion of the ophthalmic surgical instrument **801** into the eye **900** due to its thin profile and rounded edge **855**, which is less likely to bulge into and/or cause unwanted cutting at surrounding areas of the outer surface **908** of the cornea **902** near the incision **904**.

[0091] Once the ophthalmic surgical instrument **801** is introduced into the eye **900** as shown in FIG. **9B**, the ophthalmic surgical instrument **801** can be maneuvered in the anterior segment **906** of the eye **900** to interact with the capsular bag **916** of the lens **914**. For example, during cataract surgery, the ophthalmic surgical instrument **801** can be used to fragment the lens **914**, and to aspirate the fragments out of the eye **900**. Throughout the procedure, fluid can be irrigated into the eye **900** through the sleeve **820**.

[0092] FIG. **9C** shows the ophthalmic surgical instrument **801** of FIGS. **8A-8B** inserted through a capsulotomy **918** in the capsular bag **916** of the eye **900**, according to certain embodiments. In some embodiments, a laser is used to create the capsulotomy **918** before the ophthalmic surgical instrument **801** is inserted therethrough. Once the tip **850** is inserted inside the capsular bag **916**, the projection **854** may be used to help facilitate cortex removal (e.g., using the same laser or ultrasonic handpiece tip used for the capsulotomy **918**). In addition, residual lens epithelial cells may be removed by simultaneously aspirating while moving the tip **850** around inside of the capsular bag **916**, and along an inner wall of the capsular bag **916**. The projection **854** can also be used to reach under an edge of the capsulotomy **918** for a more thorough polishing inside the capsular bag **916**.

[0093] In some embodiments, for capsule polishing, the tip **850** may also comprise additional features and/or surface texture to facilitate removal of residual lens epithelial cells and sub-incisional cortex. Examples of tips with the additional features and/or the surface texture are described with reference to FIGS. **10A-10D**.

[0094] Note that one or more of the ophthalmic surgical instruments in FIGS. **10A-10D** and **11A-11B** are illustrated in a genericized manner (i.e., without additional components). For example, a sleeve is typically used with one or more of the ophthalmic surgical instruments in FIGS. **10A-10D** and **11A-11B** (to irrigate fluid into the eye **900**); however, the sleeve and one or more other components have been omitted for visual clarity.

[0095] FIG. **10A** shows the distal end **802** of the ophthalmic surgical instrument **801** with a tip **1050A** having grooves **1070**, according to certain embodiments. The grooves **1070** are located on the edge **855** of the projection **854**. The grooves **1070** are formed by a plurality of ridges and

depressions that are equally disposed across the edge **855** of the projection **854**. As an example, each of the plurality of ridges and depressions of the grooves **1070** extend across the edge **855** by 0.002 inches to 0.020 inches. While the grooves **1070** are shown on the edge **855** of the projection **854**, alternatively or additionally, they may be placed on other areas of the projection, including the distal surface **856**. Additionally, in some embodiments, the grooves may extend further or completely around the outer edge of the distal surface **856**.

[0096] FIG. **10B** shows the distal end **802** of the ophthalmic surgical instrument **801** with a tip **1050B** having notches labeled **1072** (**1072a**, **1072b**, and **1072c**), according to certain embodiments. The notches **1072** are located on the distal surface **856** of the projection **854**. The tip **1050B** also includes one or more sub-projections labeled **1073** (**1073a** and **1073b**), which form the one or more notches **1072** therebetween. As an example, each of the notches **1072** extend outward from the distal surface **856** by 0.002 inches to 0.020 inches. While the notches **1072** are shown as having an oval shape, other shapes may also be used (e.g., rectangular, triangular, etc.). Additionally, while three notches **1072** are shown in FIG. **10B**, in other embodiments, other numbers of notches (and corresponding sub-projections therebetween) may be used. For example, 1, 2, 4, 5, 10, 15, 20, etc. notches **1072** may be used.

[0097] FIG. **10C** shows the distal end **802** of the ophthalmic surgical instrument **801** with a tip **1050C** having wings labeled **1074** (**1074a** and **1074b**), according to certain embodiments. The wings **1074** are located on the projection **854** and extend laterally outwards therefrom. As an example, the wings **1074** extend outward from the projection **854** by 0.005 inches to 0.020 inches. In the illustrated example, the wings **1074** are semi-circular, but may also be rectangular, triangular, or other similar shape. Further, while the embodiment shown in FIG. **10** shows two wings, fewer or more wings can be used (e.g., 1, 3, 4, 5, 10, etc.). The wings **1074** may also be disposed at different locations on the projection **854**.

[0098] FIG. **10D** shows the distal end **802** of the ophthalmic surgical instrument **801** with a tip **1050D** having a distal surface **1076** that is textured, the textured distal surface **1076** including a portion of the projection **854**, according to certain embodiments. The textured distal surface **1076** has a roughened surface texture which may be etched (e.g., with a laser), abraded (e.g., with sandpaper), and/or coated (e.g., a tacky coating) onto the surface. As an example, the roughened surface texture of the distal surface **1076** has a roughness with an Ra between 0.012 micrometers (μm) to 25 micrometers.

[0099] Note that the features (e.g., grooves **1070**, notches **1072**, wings **1074**, and textured surface **1076**) shown in FIGS. **10A-10D** are each shown on different tips, but that one tip may include any combination of two or more of such features. For example, a tip may include the grooves **1070** and the notches **1072**, the notches **1072** and the wings **1074**, or the wings **1074** and the textured surface **1076**. Further, the features described with reference to FIGS. **10A-10D** may also be implemented with the projection **754** of the sleeve **720** shown in FIGS. **7A-7B**.

[0100] The grooves **1070**, the notches **1072**, the wings **1074**, and the textured surface **1076** shown in FIGS. **10A-10D** each provide additional scraping edges for cortical material removal (especially under the surface of a capsulotomy). For example, during capsule polishing, these features help with the removal of residual lens epithelial cells when the tip (e.g., tip **1050A-D**) is slid against a capsule wall. The residual lens epithelial cells removed from the capsule wall are then aspirated through the center opening **858**.

[0101] FIG. **11A** shows a distal end **1102** of an ophthalmic surgical instrument **1101** with a tip **1150A**, according to certain embodiments. The tip **1150A** is a phacoemulsification tip configured for ultrasonic removal of lens material without an optical fiber. The tip **1150A** includes cylindrical hub **1152**, projection **1154**, distal surface **1156**, center opening **1158**, and body **1160** similar to the tip **850** shown in FIGS. **8A-8B** but does not include the slot **862** for the optical fiber **240**.

[0102] FIG. **11B** shows the distal end **1102** of the ophthalmic surgical instrument **1101** with another tip **1150B**, according to certain embodiments. The tip **1150B** is another example of a

phacoemulsification tip without the optical fiber **240**, which may be used on a hybrid phacoemulsification needle. An example hybrid phacoemulsification needle is described in U.S. Pat. No. 11,185,442 titled “Hybrid Phacoemulsification Needle” filed May 8, 2019 which is hereby incorporated by reference in its entirety as though fully and completely set forth herein.

[0103] The tip **1150B** includes an angled distal surface **1180** around the center opening **1158** with the projection **1154** and the flat distal surface **1156** extending outwards therefrom. Additionally, in the illustrated example, the ophthalmic surgical instrument **1101** includes a shaft **1110** having a curved or bent portion along a length thereof; however, in some other embodiments, the shaft **1110** may also be substantially straight (without the curved portion).

[0104] In one example of the hybrid phacoemulsification needle with the tip **1150B**, the shaft **1110** has an interior surface, an exterior surface, and a distal end terminating in a distal edge. The shaft **1110** further has a central bore extending therethrough. The central bore is defined by the interior surface of the hollow shaft **1110**; and the tip **1150B** may include a first over mold located on the exterior surface and distal edge of the hollow shaft **1110**, the first over mold covering at least a portion of a periphery of the exterior surface of the hollow shaft **1110**, the first over mold covering the distal edge and terminating at the central bore.

[0105] In some other examples of the hybrid phacoemulsification needle with tip **1150B**, the shaft **1110** comprises a through hole extending from the exterior surface to the interior surface of the shaft **1110**, where the first over mold substantially fills the through hole. The first over mold may have a rounded front edge located over the distal edge of the hollow shaft **1110** and a rounded trailing edge located on the exterior surface of the hollow shaft **1110**. The first over mold may extend circumferentially around the entire perimeter of the exterior surface of the hollow shaft **1110**. The first over mold may be made of a polymer.

[0106] In yet some other examples of the hybrid phacoemulsification needle with tip **1150B**, the phacoemulsification needle further comprises a second over mold located on the first over mold, the second over mold covering at least a portion of a periphery of an exterior surface of the first over mold, the second over mold covering the rounded front edge of the first over mold and terminating at the central bore. The second over mold may be embedded in the first over mold such that the exterior surface of the first over mold is continuous with an exterior surface of the second over mold. The second over mold may extend circumferentially around the entire perimeter of the exterior surface of the hollow shaft. The second over mold may be made of silicone.

[0107] In various embodiments, the projections (e.g., projections **454**, **854**, and **1154**) and their various features (e.g., grooves **1070**, notches **1072**, and/or wings **1074**) may be injection molded or welded to the tip (e.g., tip **450**, **850**, **1050A-D**, and/or **1150A-B**). For example, in some embodiments, the tip may be made of plastic (e.g., elastomer, high density thermoplastic elastomer, thermoplastic polyurethane, etc.) and injection molded over the distal end **212** of the shaft **210**. In some embodiments, the tip may be made of metal and may be laser welded onto the distal end **212** of the shaft **210**. Other materials (e.g., ceramics, alumina, etc.) and formation/attachment methods may also be used.

[0108] FIG. **12** illustrates a method **1200** for performing an ophthalmic surgical procedure (e.g., a phacoemulsification procedure), according to an embodiment. The elements provided in the flowchart are illustrative only. Various provided elements may be omitted, additional elements may be added, and/or various elements may be performed in a different order than provided below.

[0109] At block **1202**, a distal end (e.g., distal end **802**) of an ophthalmic surgical instrument (e.g., ophthalmic surgical instrument **801**) may be inserted into an eye (e.g., eye **900**) of a patient toward a treatment area. For example, the operator positions the distal end **802** of the ophthalmic surgical instrument **801** adjacent the treatment area such that the distal end **802** of the optical fiber **240** faces the treatment area (e.g., a lens of an eye) and such that the distal surface **856** of the projection **854** faces the treatment area.

[0110] At block **1204**, a phacoemulsification procedure may be performed with the ophthalmic

surgical instrument in the eye. For example, the phacoemulsification procedure can be performed with the ophthalmic surgical instrument **801** inserted in the eye **900**. In some embodiments, during the phacoemulsification procedure, the operator delivers laser energy through an optical fiber (e.g., optical fiber **240**) to the treatment area.

[0111] At block **1206**, without withdrawing the distal end of the ophthalmic surgical instrument from the eye, cortex removal and capsule polishing may be performed with the projection.

[0112] In some embodiments, the method **1200** may further include reaching under a capsulotomy to facilitate removal of cortical material, including sub-incisional cortex. In some embodiments, the projection **854** may also be used as a hook to gently pull on the capsulotomy to further extend the projection's reach for removal of additional cortical material.

[0113] The present disclosure may be embodied in other specific forms without departing from its spirit or essential characteristics. The described embodiments are to be considered in all respects only as illustrative and not restrictive. The scope of the present disclosure is, therefore, indicated by the appended Claims rather than by this Detailed Description. All changes which come within the meaning and range of equivalency of the Claims are to be embraced within their scope.

[0114] Reference throughout this specification to features, advantages, or similar language does not imply that all the features and advantages that may be realized with the present disclosure should be or are in any single embodiment of the disclosure. Rather, language referring to the features and advantages is understood to mean that a specific feature, advantage, or characteristic described in connection with an embodiment is included in at least one embodiment of the present disclosure. Thus, discussions of the features and advantages, and similar language, throughout this specification may, but do not necessarily, refer to the same embodiment.

[0115] Furthermore, the described features, advantages, and characteristics of the disclosure may be combined in any suitable manner in one or more embodiments. One skilled in the relevant art will recognize, in light of the description herein, that the disclosure can be practiced without one or more of the specific features or advantages of a particular embodiment. In other instances, additional features and advantages may be recognized in certain embodiments that may not be present in all embodiments of the present disclosure.

[0116] Reference throughout this specification to “one embodiment”, “an embodiment”, or similar language means that a particular feature, structure, or characteristic described in connection with the indicated embodiment is included in at least one embodiment of the present disclosure. Thus, the phrases “in one embodiment”, “in an embodiment”, and similar language throughout this specification may, but do not necessarily, all refer to the same embodiment.

[0117] The foregoing description is provided to enable any person skilled in the art to practice the various embodiments described herein. Various modifications to these embodiments will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other embodiments. Thus, the claims are not intended to be limited to the embodiments shown herein but are to be accorded the full scope consistent with the language of the claims.

Claims

1. An ophthalmic surgical instrument comprising: a shaft having a distal end configured to be inserted into an eye of a patient toward a treatment area; and a projection located at the distal end of the shaft, the projection having a distal surface that extends at an angle relative to a longitudinal axis of the shaft, the distal surface configured to be positioned facing the treatment area during use of the ophthalmic surgical instrument.
2. The ophthalmic surgical instrument of claim 1, wherein the projection further comprises grooves on an edge of the projection.
3. The ophthalmic surgical instrument of claim 1, wherein the projection further comprises one or more notches on the projection.

- 4.** The ophthalmic surgical instrument of claim 1, wherein the projection further comprises at least one wing extending out of the projection.
 - 5.** The ophthalmic surgical instrument of claim 1, wherein the distal surface of the projection is a textured surface for polishing a lens capsule of the eye.
 - 6.** The ophthalmic surgical instrument of claim 1, further comprising a tip coupled to the shaft, wherein the projection is located on the tip.
 - 7.** The ophthalmic surgical instrument of claim 6, wherein the tip comprises: an aspiration channel providing a pathway through the tip to the shaft, wherein material aspirated at the treatment area flows through the aspiration channel.
 - 8.** The ophthalmic surgical instrument of claim 7, wherein the aspiration channel is angled relative to the longitudinal axis of the shaft.
 - 9.** The ophthalmic surgical instrument of claim 1, wherein a surface area of the distal surface of the projection is at least two to eight times greater than a surface area of an end face of the shaft.
 - 10.** The ophthalmic surgical instrument of claim 1, further comprising an optical fiber for delivering laser energy to the treatment area, the optical fiber having a distal end configured to be positioned facing the treatment area during use of the ophthalmic surgical instrument.
 - 11.** The ophthalmic surgical instrument of claim 10, further comprising: a tip coupled to the shaft, wherein the projection is located on the tip; an aspiration channel providing a pathway through the tip to the shaft, wherein material aspirated at the treatment area flows through the aspiration channel; wherein the optical fiber is disposed outside and adjacent the shaft, and wherein the optical fiber enters the aspiration channel at the tip.
 - 12.** The ophthalmic surgical instrument of claim 1, further comprising an irrigation sleeve around the shaft, and wherein the projection is located on the irrigation sleeve.
 - 13.** The ophthalmic surgical instrument of claim 12, wherein the irrigation sleeve comprises a proximal hub configured to be coupled to a housing of the ophthalmic surgical instrument and a distal tube configured to fit around the shaft.
 - 14.** The ophthalmic surgical instrument of claim 1, wherein the shaft is tubular, and wherein the tubular shaft is configured for aspiration from the treatment area.
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